

The Proof Room

A Chapter 2 escape room • 2.4 – 2.6 • teams of 3 or 4 • one packet per team

The door to Room 406 has four locks, and every lock is a proof. Each lock you open hands your team one digit of the escape code. The locks check themselves — you will know the moment one opens, and the moment one doesn't. No one leaves until the door does.

HOW A LOCK WORKS

1. Read the Given and the Prove. Build the proof in the ledger — but every statement must come from the STATEMENT BANK and every reason from the REASON CODES.
2. On each line write the statement's bank number in S #, the statement itself, the reason's code in Code, and the reason's name.
3. STATEMENT LOCK: multiply each line number by its S #, then add. REASON LOCK: multiply each line number by its Code, then add.
4. Both totals must match the numbers printed on the page. If either misses, something is wrong — a trap statement, a wrong reason, or lines out of order. Find it.
5. When both match, record your escape digit and move on. Four digits open the door.

ROLES — rotate one seat every lock

READER — reads the Given and Prove aloud; owns the banks; says “that one is a trap” when it is.

SCRIBE — the only pen on the ledger.

CHECKER — runs the lock arithmetic. Twice.

CAPTAIN — settles arguments, records the digit, signs the door.

HOUSE RULES

Line 1 is the given, word for word. The last line is exactly what you were asked to prove.

A reason is a given, a definition, a property, a postulate, or a theorem. A theorem is a reason only AFTER it has been proved.

Every bank has traps. The bank also decides the path: if the statement you wanted isn't there, take the route that is.

REASON CODE BANK — every reason in the room, with its code. Each lock repeats the ones it needs.

98 Addition Property of Equality	27 Division Property of Equality
45 Angle Addition Postulate	44 Given
13 Congruent Complements Theorem	67 Linear Pair Postulate
87 Congruent Supplements Theorem	31 Multiplication Property of Equality
66 Definition of angle bisector	64 Reflexive Property of Equality
38 Definition of complementary angles	94 Right Angles Congruence Theorem
42 Definition of congruent angles	15 Segment Addition Postulate
26 Definition of linear pair (from the diagram)	81 Substitution Property of Equality
11 Definition of midpoint	43 Subtraction & Addition POE
93 Definition of perpendicular	96 Subtraction Property of Equality
14 Definition of right angle	99 Symmetric Property of Equality
24 Definition of supplementary angles	19 Transitive Property of Equality
77 Distributive Property	88 Vertical Angles Congruence Theorem

Lock 1 — Justify each step (2.4)

Given: $4(x + 2) = 2x + 20$ Prove: $x = 6$

Statement bank — more statements than lines. The extras are traps.

1. $2x + 8 = 20$	5. $6x + 8 = 20$
2. $2x = 28$	6. $4x + 8 = 2x + 20$
3. $x = 6$	7. $4x + 2 = 2x + 20$
4. $4(x + 2) = 2x + 20$	8. $2x = 12$

Reason codes for this lock

- 98 Addition Property of Equality
 77 Distributive Property
 27 Division Property of Equality
- 44 Given
 31 Multiplication Property of Equality
 96 Subtraction Property of Equality

#	S #	Statement	Code	Reason
1				
2				
3				
4				
5				

STATEMENT LOCK $1 \times \underline{\hspace{1cm}} + 2 \times \underline{\hspace{1cm}} + 3 \times \underline{\hspace{1cm}} + 4 \times \underline{\hspace{1cm}} + 5 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ **must equal 66**
REASON LOCK $1 \times \underline{\hspace{1cm}} + 2 \times \underline{\hspace{1cm}} + 3 \times \underline{\hspace{1cm}} + 4 \times \underline{\hspace{1cm}} + 5 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ **must equal 1005**
Both locks match? Your escape digit for this lock: Lock 1: the value of $x \rightarrow \underline{\hspace{1cm}}$

Lock 2 — Angle bisector (2.5)

Given: Ray BD bisects $\angle ABC$; $m\angle ABD = (7x - 4)^\circ$, $m\angle DBC = (4x + 17)^\circ$ Prove: $x = 7$

Statement bank — more statements than lines. The extras are traps.

1. $3x = 21$

2. $3x = 13$

3. $\angle ABD \cong \angle ABC$

4. $m\angle ABD + m\angle DBC = 180^\circ$

5. $\angle ABD \cong \angle DBC$

6. $7x - 4 = 4x + 17$

7. $x = 7$

8. $m\angle ABD = m\angle DBC$

9. Ray BD bisects $\angle ABC$

Reason codes for this lock

- 45 Angle Addition Postulate

66 Definition of angle bisector

42 Definition of congruent angles

11 Definition of midpoint

27 Division Property of Equality

44 Given

81 Substitution Property of Equality

43 Subtraction & Addition POE

#	S #	Statement	Code	Reason
1				
2				
3				
4				
5				
6				

STATEMENT LOCK $1 \times \underline{\hspace{1cm}} + 2 \times \underline{\hspace{1cm}} + 3 \times \underline{\hspace{1cm}} + 4 \times \underline{\hspace{1cm}} + 5 \times \underline{\hspace{1cm}} + 6 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ **must equal 114**

REASON LOCK $1 \times \underline{\hspace{1cm}} + 2 \times \underline{\hspace{1cm}} + 3 \times \underline{\hspace{1cm}} + 4 \times \underline{\hspace{1cm}} + 5 \times \underline{\hspace{1cm}} + 6 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ **must equal 1003**

Both locks match? Your escape digit for this lock: Lock 2: the value of $x \rightarrow \underline{\hspace{1cm}}$

Lock 3 — Prove the Congruent Complements Theorem (2.6)

Given: $\angle 1$ and $\angle 2$ are complementary; $\angle 2$ and $\angle 3$ are complementary Prove: $\angle 1 \cong \angle 3$

Statement bank — more statements than lines. The extras are traps.

1. $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$

2. $m\angle 2 = m\angle 2$

3. $m\angle 1 = m\angle 2$

4. $m\angle 1 = m\angle 3$
5. $m\angle 1 + m\angle 2 = 180^\circ$; $m\angle 2 + m\angle 3 = 180^\circ$

6. $m\angle 1 + m\angle 2 = 90^\circ$; $m\angle 2 + m\angle 3 = 90^\circ$

7. $\angle 1 \cong \angle 3$

8. $\angle 1$ and $\angle 2$ are complementary; $\angle 2$ and $\angle 3$ are complementary

Reason codes for this lock

- 13 Congruent Complements Theorem

38 Definition of complementary angles

42 Definition of congruent angles

24 Definition of supplementary angles

44 Given
- 64 Reflexive Property of Equality

81 Substitution Property of Equality

96 Subtraction Property of Equality

99 Symmetric Property of Equality

#	S #	Statement	Code	Reason
1				
2				
3				
4				
5				
6				

STATEMENT LOCK $1 \times \underline{\hspace{1cm}} + 2 \times \underline{\hspace{1cm}} + 3 \times \underline{\hspace{1cm}} + 4 \times \underline{\hspace{1cm}} + 5 \times \underline{\hspace{1cm}} + 6 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ **must equal 93**

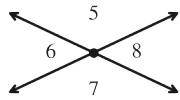
REASON LOCK $1 \times \underline{\hspace{1cm}} + 2 \times \underline{\hspace{1cm}} + 3 \times \underline{\hspace{1cm}} + 4 \times \underline{\hspace{1cm}} + 5 \times \underline{\hspace{1cm}} + 6 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ **must equal 1351**

Both locks match? Your escape digit for this lock: Lock 3: the line number where the shared angle disappears →

Lock 4 — The sabotaged proofs (2.6)

Each proof below is complete and every statement is correct. In each one, exactly ONE reason is wrong. Find the line, write the reason that belongs there, and say in one sentence why the sabotage fails.

Proof A — the Vertical Angles Theorem



Given: $\angle 5$ and $\angle 7$ are vertical angles Prove: $\angle 5 \cong \angle 7$

Statements	Reasons
1. $\angle 5$ and $\angle 7$ are vertical angles	1. Given
2. $\angle 5$ and $\angle 6$ form a linear pair; $\angle 6$ and $\angle 7$ form a linear pair	2. Definition of linear pair (from the diagram)
3. $\angle 5$ and $\angle 6$ are supplementary; $\angle 6$ and $\angle 7$ are supplementary	3. Definition of supplementary angles
4. $\angle 5 \cong \angle 7$	4. Congruent Supplements Theorem

Sabotaged line: ____ **Correct reason:** _____ **code** ____
Why it fails:

Proof B — the Congruent Supplements Theorem

Given: $\angle 1$ and $\angle 2$ are supplementary; $\angle 2$ and $\angle 3$ are supplementary Prove: $\angle 1 \cong \angle 3$

Statements	Reasons
1. $\angle 1$ and $\angle 2$ are supplementary; $\angle 2$ and $\angle 3$ are supplementary	1. Given
2. $m\angle 1 + m\angle 2 = 180^\circ$; $m\angle 2 + m\angle 3 = 180^\circ$	2. Definition of supplementary angles
3. $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$	3. Substitution Property of Equality
4. $m\angle 2 = m\angle 2$	4. Reflexive Property of Equality
5. $m\angle 1 = m\angle 3$	5. Subtraction Property of Equality
6. $\angle 1 \cong \angle 3$	6. Congruent Supplements Theorem

Sabotaged line: ____ **Correct reason:** _____ **code** ____
Why it fails:

LOCK 4 CHECK

Look up the two CORRECT reasons in the code bank on page 1. Their codes add to 109.
Your escape digit for this lock: the sabotaged line in Proof B.

THE DOOR

Write the four escape digits in order and bring the packet to Mr. Nappier. If the code opens the door, your team is out.

Captain: — — — — —
Time out: — — — — —